

Regulatory mechanisms of fruit ripening: an insight from DNA, histone, and RNA modifications

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Abstract

Complex regulatory networks formed by epigenetic modifications such as DNA, histones, and RNA can influence gene expression without altering DNA sequences, participating in ripening processes and quality changes in fruits. Although significant progress has been made in understanding epigenetic modifications and their roles in fruit ripening, the regulatory network underlying these complex relationships remains incompletely elucidated. Here, we summarize the key molecular regulatory networks involving DNA, histone, and RNA modifications in fruit ripening and secondary metabolite changes in recent years; review the research applications of epigenetic modification inhibitors and omics technologies, elucidating their advantages and limitations; and ultimately propose a comprehensive strategic model for future studies on epigenetic modification regulation of fruit ripening. These findings not only deepen our understanding of epigenetic modifications during fruit ripening, but also raise new questions and research directions. It provides a deeper understanding of the ripening process and quality changes in fruits, offering insights for new strategies in fruit production and integrated regulation.

Introduction

The transition to sustainable diets is one of the most critical global challenges of the 21st century. This implies an increase in the demand for horticultural plants, particularly fruit and vegetables, to replace traditional dietary patterns [1, 2]. Moreover, with societal development and progress, people's demand for these horticultural plants has also begun to shift, particularly for fruits. The demand has transitioned from "accessibility" to "high quality" and "health benefits" [3-5]. To meet market demands, systematic research has been conducted on the ripening process and quality changes of fruits. Integrated regulatory strategies encompassing fruit ripening and development [6], post-harvest storage [7], processing and transportation [8] have been developed as a result. Fruit ripening is a complex process that directly affects post-harvest yield and quality. Horticultural plants primarily feature fleshy fruits, which can be classified into two types based on their respiratory activity in the early stages of ripening: climacteric fruits and non-climacteric fruits. Climacteric fruits utilise ethylene as the core signal, exhibiting significantly enhanced respiration during early ripening [9], followed by changes in colour, aroma, texture and nutrient content [10]. In contrast, non-climacteric fruits are regulated by multiple hormones, including abscisic acid (ABA), jasmonic acid (JA) and auxin (IAA), showing minimal respiratory changes during early ripening [11, 12]. It is also primarily related to the fact that hormones can influence the expression of genes associated

with fruit ripening through transcriptional and post-translational modifications (PTM) regulation. In addition, fruit ripening is also influenced by epigenetic modifications [6, 13].

In eukaryotes, epigenetic modifications carry heritable regulatory information without altering the DNA sequence. To explain the phenomenon where biological phenotypes exhibit marked differences despite unchanged DNA sequences, scientists proposed a concept called "epigenetics" in 1942. This is a combination of "genetics" and "epigenesis" [14]. After almost 100 years, a key regulatory network has been revealed, including epigenetic modifications like DNA methylation, histone modifications, RNA modifications, changes to the structure of chromosomes, and the organisation of chromosomes into domains (TADs) [15]. In plants, studies of epigenetic modifications have been conducted in model plants like *Arabidopsis thaliana* and major crops such as *Oryza sativa*, *Zea mays*, and *Triticum aestivum* [16]. Research areas include DNA methylation [17], histone modifications [18], histone variants [19], chromatin remodelling factors [20], non-coding RNAs [21], RNA modifications [22], and chromatin higher-order structure [23]. These epigenetic modification studies also help address biological questions related to embryonic development [24], plant regeneration [25], vernalization [26], plant architecture regulation [27], nodulation and nitrogen fixation [21], biotic/abiotic stress [28-29], and hybrid vigor [30]. Research on epigenetic modifications in horticultural plants has passed the exploratory stage and entered a period of rapid development. In addition to the extensive research of epigenetic modifications in other plant species, recent years have seen analyses of developmental, ripening and secondary metabolite processes in horticultural plants. Research areas include DNA methylation and active demethylation [31], histone modifications [32], and RNA modifications [33]. Among these, climacteric fruits *Solanum lycopersicum* and non-climacteric fruits strawberry (*Fragaria × ananassa* and *Fragaria vesca*) have been the most extensively studied as classic models. Their research has driven advances in the understanding of epigenetic molecular regulatory mechanisms in fruit ripening and quality changes. Previous studies have provided detailed and accurate reviews of epigenetic modifications during fruit ripening and quality changes. These studies have advanced our understanding of fruit maturation and its underlying regulatory mechanisms [31-35]. However, the specific molecular regulatory networks involved remain poorly defined, and the potential applications of these modifications in production have yet to be fully explored. In this review, we systematically mapped the regulatory networks of key research advances in DNA, histone and RNA modifications during fruit ripening in recent years; summarised the effects of exogenous inhibitors

associated with epigenetic modifications, and reviewed the application of various omics technologies. This review aims to establish an integrated research model for epigenetic modifications that combines omics technology applications, molecular regulation analysis and exogenous substance regulation. This approach will provide a new insight and understanding of strategies for the comprehensive regulation of fruit ripening and quality changes.

DNA methylation and active demethylation

DNA methylation is widely present in eukaryotes and represents a conserved epigenetic mark. It primarily involves two types: the conversion of adenine to N^6 -methyladenine (6mA) and the conversion of cytosine to C^5 -methylcytosine (5mC) [36]. In plants, current research primarily focuses on 5mC DNA methylation. In fact, 5mC levels in plants are typically higher than those in animals, and the rate of cytosine methylation can be up to 50% [37]. However, only in *Arabidopsis thaliana* [38], *Oryza sativa* [39] and *Zea mays* [40] has 6mA methylation been found to play a crucial role in biological functions such as seed formation, vegetative growth, photosynthesis and the response to heat stress. The primary sites of DNA methylation in plants are symmetric CG, CHG, and asymmetric CHH. CG and CHG methylation simultaneously modify the corresponding cytosines in double-stranded DNA, whereas CHH methylation occurs only in single-stranded DNA [16]. Among these, chromatin methyltransferases (CMTs) that maintain CHG methylation are only found in plants. This result means that CHG methylation is much more common in plants than animals [41]. Extensive research has confirmed that *de novo* methylation in plants occurs via the RNA-directed DNA methylation (RdDM) pathway [42]. Firstly, double-stranded RNA (dsRNA) synthesised by RNA-dependent RNA polymerase 2 (RDR2) employing Pol IV transcripts as templates is processed by Dicer-like 3 (DCL3) to yield 24-nt small interfering RNA (siRNA). This siRNA is then loaded onto Argonaute 4 (AGO4), which facilitates its entry into the nucleus. Domains rearranged methyltransferase 2 (DRM2) acts in concert with AGO4 to establish DNA methylation of the Pol V transcription template strand [43–45]. Based on this RdDM pathway prevalent in plants, we mapped the DNA modification regulatory network during fruit ripening and distinguished differences between climacteric fruits and non-climacteric fruits within the regulatory network (Fig. 1).

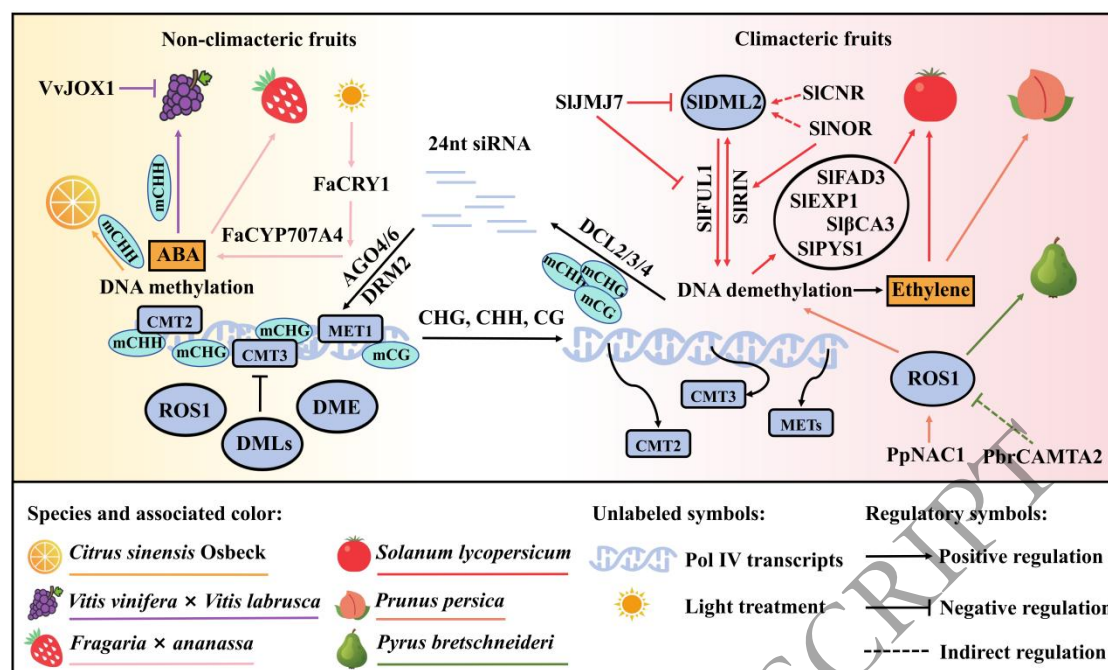


Figure 1. Regulation of RdDM pathway during fruit ripening, including both non-climacteric fruits and climacteric fruits. Non-climacteric fruits: strawberry (*Fragaria* × *ananassa*), grape (*Vitis vinifera* × *Vitis labrusca*), citrus (*Citrus sinensis* Osbeck); climacteric fruits: tomato (*Solanum lycopersicum*), peach (*Prunus persica*), pear (*Pyrus bretschneideri*). Non-climacteric fruits primarily promote ripening through DNA methylation, which might be associated with high levels of mCHH methylation and the ABA signaling pathway. Climacteric fruits primarily promote ripening through active DNA demethylation mediated by ROS1 and DML2; it is primarily linked to the impact of ethylene biosynthesis. Regulatory mechanisms across species are depicted with different colored lines. Positive regulation is indicated by arrows, while negative regulation is represented by T-shaped symbols.

High levels of DNA methylation promote the ripening of non-climacteric fruits. The maintenance of DNA methylation in plants aids the restoration of 5mC modifications in newly formed DNA strands, primarily through the action of Methyltransferase 1 (MET1) and Chromomethylase 2/3 (CMT2/3) [46]. Among these, MET1 primarily interacts with mCG-binding proteins (VIM1/2/3) to maintain mCG methylation through a semi-conservative mechanism [46]. CMT2 and CMT3 share a similar catalytic mechanism, and difference is that CMT2 primarily maintains mCHH methylation in heterochromatic regions, whereas CMT3 positively regulates mCHG methylation levels [47]. During the ripening of non-climacteric fruits, higher levels of DNA methylation are observed. This increasing trend may represent an interplay between environmental factors, plant hormones and epigenetic modifications. In addition, mCHH may be a key site that

positively regulates DNA methylation levels in non-climacteric fruits. Firstly, the ripening of non-climacteric fruits is influenced by external environmental factors, such as light and temperature. DNA methylation can act as a mechanism to respond to environmental signals, which enables fruits to adjust their ripening process in response to changes in external conditions. For example, a study of *Fragaria* × *ananassa* revealed that bagged fruits exhibit delayed ripening and significantly reduced DNA methylation levels in the promoter region of ABA 8'-hydroxylase (*FaCYP707A4*), a gene derived from *Fragaria vesca* [48], and suppressed expression of blue light receptor *CRYPTOCHROME 1* (*FaCRY1*). When exposed to light, *FaCRY1* interacted with FaAGO4, which enhanced methylation levels by binding to the *FaCYP707A4* promoter region and promoted fruit ripening [49]. Secondly, the ripening of non-climacteric fruits is regulated by multiple hormones. These hormonal signalling pathways are crucial for DNA methylation, especially in response to changes in ABA levels. The exogenous application of 500 ppm ABA + 0.1% Tween 20 to *Vitis vinifera* significantly promoted fruit ripening. Differentially expressed genes (DEGs) enriched in defence-related metabolism, chlorophyll degradation processes and ripening-related metabolic pathways, accompanied by genome-wide high levels of DNA methylation [50]. High methylation in *Citrus sinensis* Osbeck was highly correlated with the active expression of hundreds of genes, primarily involved in fruit ripening and mainly participating in the ABA hormone signalling pathway [51]. The bagging treatment reduced DNA methylation levels of *FaCYP707A4* in *Fragaria* × *ananassa*, which was also associated with reduced endogenous ABA accumulation [49]. Furthermore, significant changes in mCHH might primarily influence DNA methylation levels in non-climacteric fruits. For example, the early-ripening bud sport variety 'Fengzao' exhibited higher methylation levels in transposons and gene flanking regions than 'Kyoho' in *Vitis vinifera*, primarily due to the regulation of CHH site hypermethylation mediated by JASMONIC ACID OXIDASE 1 (*VvJOX1*) [52]. Similar findings were observed in *Vitis vinifera* varieties 'Wink' and 'Cabernet Savior', where global genome-wide DNA hypermethylation occurred during fruit ripening, primarily characterized by elevated methylation levels in CHH sequence contexts, particularly in promoter and transposon regions [53]. During the ripening process of *Citrus sinensis* Osbeck fruit, increased mCHH methylation levels led to elevated genome-wide DNA methylation levels, which are significantly higher than the increases observed in mCG and mCHG methylation levels [51]. However, high levels of whole-genome DNA methylation do not necessarily occur during the ripening process of non-climacteric fruits. The study in *Fragaria* × *ananassa* revealed that DNA

methylation generally decreased during fruit ripening, similar to the climacteric fruit *Solanum lycopersicum*. This was not driven by the regulation of DNA demethylase genes, unlike the case in *Solanum lycopersicum*, but rather by the downregulation of genes involved in maintaining DNA methylation [54]. Genome-wide DNA methylation levels in *Fragaria × ananassa* have been found to be similarly up-regulated in dark conditions, which corresponds to delayed ripening processes [49]. This suggests that the ripening of non-climacteric fruits does not rely solely on genome-wide DNA methylation, but actually involves a systematic and complex interplay between environmental factors, hormones, gene expression, and epigenetic modifications.

DNA demethylation contributes to faster ripening of climacteric fruits. There are two pathways through which DNA demethylation can occur: one is passive demethylation, which is replication-dependent and involves the gradual dilution of DNA methylation due to its failure to be maintained. The other pathway is active demethylation, which is non-replication-dependent and requires enzyme catalysis [55]. This represents a crucial dynamic equilibrium process within living organisms. In plants, active demethylation is primarily regulated by the combined action of Repressor of Silencing 1 (ROS1), Demeter (DME) and Demeter-like proteins (DML), which replace 5mC with unmethylated cytosine [37, 56, 57]. Unlike non-climacteric fruits, climacteric fruits undergo active DNA demethylation during ripening, which is primarily influenced by ethylene signalling. For example, during the ripening process of *Prunus persica*, the transcription level of NAC domain-containing protein 1 (PpNAC1) increased, directly activates the expression of multiple ethylene signalling-associated genes and initiated the process of active DNA demethylation through PpDML1 (also known as PpROS1) [58]. Similar findings were observed in *Solanum lycopersicum*, where active DNA demethylation positively regulates fruit ripening by controlling ethylene biosynthesis and carotenoid accumulation [59, 60]. At the molecular regulatory level, DML2 and ROS1 play a key role in active DNA demethylation in climacteric fruits. The role of DML2 in demethylation during the ripening and quality development of *Solanum lycopersicum* fruit was considered a landmark discovery [31]. Specifically, SIDML2 serves as a pivotal regulator of active demethylation during fruit ripening [61, 62], positively regulating Ripening Inhibitor (SIRIN) and FRUITFULL 1 (SIFUL1) to influence chromatin accessibility [59]. During this process, JUMONJI DOMAIN-CONTAINING PROTEIN 7 (SIJM7) negatively regulates SIDML2 and SIFUL1 by suppressing the expression of ethylene biosynthetic and ripening-related genes via Histone H3 lysine 4 trimethylation (H3K4me3) [60]. NON-RIPENING (SINOR) positively regulates SIDML2

and SIRIN [61]. SlbCA3 and SIFAD3, which encode b-carbonic anhydrase and x-3 fatty acid desaturase respectively, are activated at the transcriptional level through SIDML2-mediated DNA demethylation. Both genes participate in the regulation of ripening and disease resistance in fruit [63]. However, when SIRIN expression was restored in *Sldml2* mutants (in which both *SIRIN* and *SIFUL1* were silenced), the expression of more than half of DEGs failed to recover. These results suggest the presence of additional unknown factors downstream of SIDML2 that regulate fruit ripening [59]. Additionally, recent studies have revealed that ROS1 plays a crucial role in climacteric fruits. Analysis of 41 germplasm resources showed that *Pyrus bretschneideri* exhibited an overall increase in DNA methylation during domestication and improvement. This increase in methylation correlates with downregulation of PbrROS1 and primary positive regulation via CALMODULIN-BINDING TRANSCRIPTION ACTIVATOR 2 (PbrCAMTA2) [64]. High levels of DNA methylation caused a reduction in the accumulation of lutein, β -carotene and ABA in *Pyrus bretschneideri*, and delayed fruit ripening [65]. Overall, DML2 and ROS1 are key nodes in active demethylation, jointly regulating the ripening process of climacteric fruits via ethylene signalling. However, there may still be unknown regulatory genes downstream of DML2, the precise crosstalk and regulatory relationship between active demethylation and ethylene signalling pathways remain unclear. In fact, genome-wide methylation level analysis has also been completed in *Malus domestica*, but no detailed functional analysis has been performed on differentially methylated genes exhibiting differential methylation levels [66].

Active DNA demethylation positively regulates the biosynthesis of secondary metabolites during fruit ripening. The synthesis and degradation of these secondary metabolites are jointly regulated by internal development clues and various other factors [32]. Unlike the differences in DNA modification between climacteric and non-climacteric fruits during ripening, both types exhibit active DNA demethylation associated with anthocyanin biosynthesis. This change is primarily mediated by MYB domain proteins (MYB) [66], CHALCONE ISOMERASE (CHI), CHALCONE SYNTHASE (CHS) [67], UDP-glucose flavonoid glycosyltransferase (UGT) [68], and FLAVANONE 3-HYDROXYLASE (F3H) [69], which regulate DNA demethylation during fruit anthocyanin biosynthesis [70]. Furthermore, a study of *Citrus limon* revealed that genes associated with the accumulation of the primary metabolite citric acid are closely linked to DNA methylation levels. These genes included MYB transcription factors (CIPH1/4/5) and ANTHOCYANIN 1 (CIAN1) [71]. This may be related to the low levels of DNA methylation

observed in the promoter region of *pleckstrin homologue 5 (PH5)* [72]. It is suggested that there may be undiscovered links between DNA modifications and changes in primary and secondary metabolites during fruit ripening. Another key discovery is that DNA modifications transform the toxic, bitter compounds in *Solanum lycopersicum* fruit to non-toxic, non-bitter compounds during ripening. During fruit ripening, active DNA demethylation plays a key role in the biosynthesis of steroidal glycoalkaloids (SGAs). DNA demethylation, which is mediated by *SIDML2*, drives SGA detoxification during domestication [73]. Variation in DNA methylation may represent the genetic basis for this metabolic diversity in plants [74]. However, research on DNA modifications related to the biosynthesis of fruit secondary metabolites is still limited. The biosynthetic pathways of flavonoids remain poorly understood, and the regulatory roles of key structural genes remain unclear.

Histones modifications

Histones are essential components of chromosomes. In eukaryotes, the highly conserved core histones H2A, H2B, H3, and H4 form the stable octameric core structure of nucleosomes, with H1 family members serving as linker histones [75]. Due to their complex structure, histones play a vital role in epigenetic regulation and gene expression control. The diverse array of modification types, sites, and variants collectively form the unique "histone code" [76]. In addition to the classic methylation modification, numerous histone post-translational modifications (HPTMs), such as acetylation and ubiquitination, can also occur on the N-terminal "tail" [13]. Currently, extensive research on histone modifications in horticultural plants has revealed that these modifications participate in crucial biological processes including growth [77], reproduction [78], biotic stress [79], and abiotic stress [80]. Here, we summarize the key regulatory networks governing histone modifications during fruit ripening and elucidate the relationships among these modifications (Fig. 2).

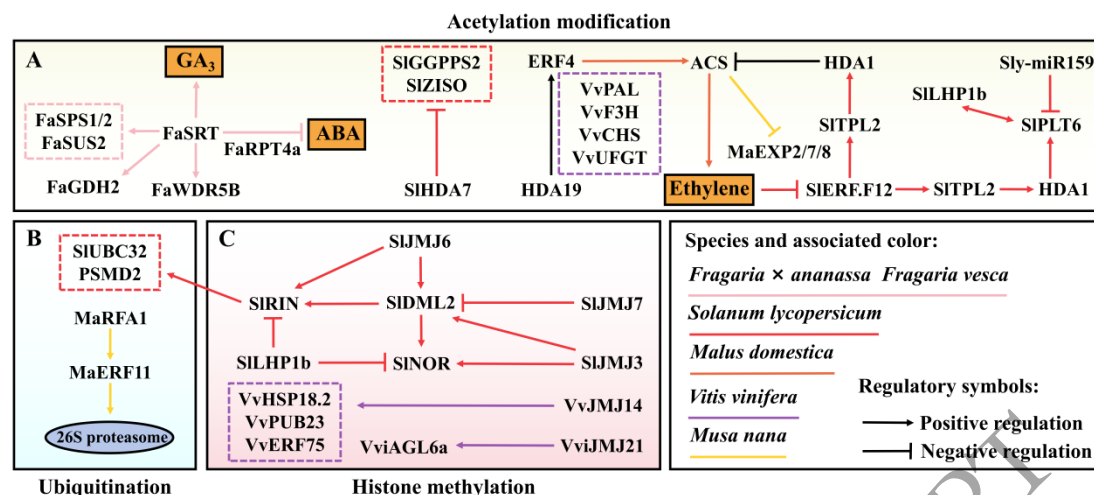


Figure 2. Regulation of histones and corresponding post-translational modifications during fruit ripening. The modifications include **(A)** acetylation, **(B)** ubiquitination, and **(C)** histone methylation. Fruits include tomato (*Solanum lycopersicum*), strawberry (*Fragaria × ananassa* and *Fragaria vesca*), grape (*Vitis vinifera*), apple (*Malus domestica*), and banana (*Musa nana*). **(A)** The 'HDA-ERF' module influences climacteric fruit ripening and histone acetylation via regulation of ethylene signaling. The histone modification-related enzyme SRT influences non-climacteric fruit ripening via participation in ABA hormone signaling pathways. **(B)** E3 ubiquitin ligase regulates the 26S proteasome pathway and undergoes ubiquitination modification during fruit ripening. **(C)** JMJ is a key enzyme influencing histone methylation during fruit ripening and is directly linked to active DNA demethylation. Black lines indicate regulatory pathways shared across multiple species. Regulatory mechanisms unique to different species are represented by lines of varying colors. Positive regulation is indicated by arrows, while negative regulation is represented by T-shaped symbols.

JMJ is a key class of enzymes that influences histone methylation modifications during fruit ripening. Histone methylation is a crucial form of epigenetic regulation. It primarily functions by regulating gene transcription, chromatin structure, and genomic stability [13]. Similar to the mechanism of DNA methylation, histone methylation is maintained at a certain level through the action of methyltransferases and demethylases [81-83]. Methylation modifications on histones H3K4, H3K36, and H3K79 are enriched in euchromatin, whereas those on H3K9, H3K27, and H4K20 typically accumulate in heterochromatin or facultative heterochromatin [84]. The ripening process of fruits is primarily regulated by the histone demethylase JMJ, which mainly affects two types of modifications: H3K4me3 and Histone H3 at lysine 27 trimethylation (H3K27me3). This histone methylation mechanism has been systematically studied in *Solanum lycopersicum* fruits and

is associated with the RdDM pathway. Specifically, SIJM6 and SIJM7 can negatively regulate H3K27me3 [85] and H3K4me3 [60], respectively. This leads to histone demethylation in fruits and subsequently promotes ripening. This modification is strongly correlated with the DNA demethylase SIDML2 and participates in the regulation of SIRIN and SINOR [61]. During this process, SIJM3 exhibits a positive correlation with SIDML2 and SINOR [86]; heterochromatin protein 1b (SIHP1b) shows a negative correlation with SIDML2 and SIRIN [35], both mediated through H3K4me3 modification. In addition, loss of SIJM10 function resulted in smaller fruit, while its overexpression significantly increased fruit size by promoting epidermal cell expansion [87]. This role of JMJ enzymes in histone methylation has also been observed in *Vitis vinifera*, where they participate in fruit ripening. For example, VviAGL6a directly activates VviJM21 expression by binding to its promoter, thereby promoting early ripening [88]. VviJM14 influences the transcriptional expression of HEAT SHOCK PROTEIN (*VvHSP18.2*), *PLANT U-BOX* (*VvPUB23*), and *ETHYLENE RESPONSE FACTOR* (*VvERF75*) by regulating the levels of H3K4me3 modification, thereby regulating fruit ripening [89]. However, studies on histone methylation modifications during fruit ripening in other species are rare.

Hormone signaling and histone deacetylases play crucial regulatory roles during fruit ripening. Histone acetylation on nucleosomes represents one of the earliest HPTMs studied in epigenetic regulation and is closely associated with gene activation [90]. Acetylation occurs on conserved lysine residues at the amino-terminal tail of core histones, with all four core histones capable of acetylation *in vivo* or *in vitro*. Generally, H3 and H4 are more readily acetylated than H2A and H2B, and each histone acetyltransferase modifies distinct lysine residues [91]. Histone deacetylase (HDA1) is a key enzyme for histone acetylation during the ripening of climacteric fruits. This pathway is primarily influenced by ethylene signalling, which negatively regulates SIERF.F12, a transcription factor containing the Ethylene-responsive element-binding factor-associated Amphiphilic Repression (EAR) motif. It suppresses the transcription of ripening-related genes by recruiting the co-repressors TOPLESS 2 (TPL2) and SIHDA1 [92]. The SIPLT6 transcription factor contains the EAR motif and recruits SIHDA1 via its C-terminal domain. It also interacts with LIKE HETEROCHROMATIN PROTEIN 1b (SIHHP1b), providing a link to this maturation-inhibiting mechanism [93]. Additionally, MaERF11 recruits HDA1 during fruit ripening to suppress expression of acetyl-CoA synthetase (ACS) family [35], and directly binds to the promoters of the ripening-associated EXPORTIN family (*MaEXP2/7/8*) to reduce acetylation levels of histone H3

and H4 in *Musa nana* [94]. Response pathways regulating ethylene signaling via the 'HDA-ERF' module have also been found in *Malus domestica* [95] and *Vitis vinifera* [96], influencing fruit ripening and anthocyanin biosynthesis. SIRTUINS (SRTs) may represent a key group of enzymes that influence histone modifications during non-climacteric fruit ripening, which is primarily regulated by ABA signalling [97]. Among these, FaSRT1-2 actively promotes vegetative growth and fruit ripening in strawberry. This is achieved by regulating ABA and gibberellin (GA) signaling pathways [98]. FaSRT2-1 primarily influences the expression of genes such as *solanese diphosphate synthase (SPS)*, *sucrose synthase (SUS)*, and *human WDR5 homolog b (WDR5B)* by participating in energy metabolism [99]. However, our understanding of this histone deacetylase remains limited, as research on it has only recently begun and has primarily been conducted in *Fragaria × ananassa*. Does the SRT family perform the same function in other non-climacteric fruits? What are the specific regulatory relationships between the SRT family and the hormones (ABA and GA)? Further research is required to answer these questions.

Protein ubiquitination occurs via the ubiquitin-proteasome pathway (UPP), where 76-amino-acid-long ubiquitin proteins are linked to lysine residues through amide bonds [100]. Histone ubiquitination primarily occurs on H2A and H2B. Histone H2A ubiquitination mainly targets lysine 119, primarily catalyzed by RING domain-containing E3 ubiquitin ligases [101]. This is a primary type of histone ubiquitination modification that affects fruit ripening. Research on ubiquitination during fruit ripening is currently limited, with only a few studies reported in *Musa nana* and *Solanum lycopersicum*. Among these, the E3 ubiquitin ligase MaRFA1 accumulates and anchors MaERF11 during fruit ripening, causing a decrease in MaERF11 protein abundance and regulating the 26S proteasome pathway [102]. In addition, RIN transcription factors can directly bind to the promoters of *ubiquitin-conjugating enzyme (SIUBC32)* and 26S proteasome regulatory subunit (PSMD2); loss of RIN function affects protein ubiquitination within nuclei and consequently participates in the regulation of fruit ripening in *Solanum lycopersicum* [103]. This suggests that there might be significant regulatory crosstalk between histone methylation and ubiquitination during fruit ripening. However, research on the important links between these different histone modifications remains limited and has been conducted in only a few fruit species. Elucidating this key regulatory pathway may provide new insights into the epigenetic modifications involved in fruit ripening.

Histone modifications play a role in the synthesis and accumulation of various secondary

metabolites during fruit ripening, including pigments, flavours, and aromatic compounds. This constitutes a complex epigenetic regulatory process. Firstly, histone modifications influence fruit color changes by regulating the dynamic balance of secondary metabolites, such as anthocyanins, carotenoids, and chlorophyll. In the synergistic regulation of DNA and histone methylation, low methylation levels induce increased expression of genes involved in photosynthesis, flavonoid biosynthesis, and hormone signaling pathways in *Solanum lycopersicum* and *Fragaria vesca* fruits [104]. During the ubiquitination process, stearoyl-ACP desaturase 8 (SISAD8) interacts with E3 ligase (SISP1) within the chloroplast-associated protein degradation pathway in chloroplasts. This disrupts the normal function of the pathway and inhibits the transition of chloroplasts to chromoplasts [105]. During the process of acetylation modification, DREB and EAR motif protein (SIDEAR1) recruited Topless 2 (SITPL2)-HDA1/HDA3 complex via its C-terminal EAR motif. This reduces the levels of histone acetylation in the SHOOT GRAVITROPISM 1 (SISGR1) promoter region and regulates chlorophyll metabolism in fruit [106]. This 'TPL-HDA1/HDA3 complex' module also reduced histone acetylation levels in lycopene biosynthesis genes [107]. Similar findings were observed in *Malus domestica* fruit, where transcription factors containing the EAR motif suppressed carotenoid biosynthesis via a cascade regulated by TPL-mediated histone deacetylation [108]. Unlike the regulatory mechanisms of other secondary metabolites, the expression of genes involved in fruit anthocyanin synthesis is primarily influenced by the inhibitory effect of histone deacetylases binding to ERF transcription factors on GCC-box [109], particularly the MYB transcription factor family [68]. Secondly, histone modifications might influence the release of aromatic compounds in fruits. A classic example is the interaction identified between PpMADS2 and E3 ubiquitin ligase CONSTITUTIVE PHOTOMORPHOGENIC 1 (PpCOP1) in *Prunus persica* L. Batsch. This module regulated linalool biosynthesis by activating the expression of PpTPS1, a terpene synthase [110]. This phenomenon of aroma substance release may also be regulated during DNA and histone methylation processes [104]. In addition, SIDML2-mediated DNA demethylation regulated the SGA detoxification process in *Solanum lycopersicum* fruit, transforming it into a non-toxic and non-bitter state, which was a significant advancement in recent studies of fruit epigenetic modification [73]. This process is primarily regulated by the GLYCOALKALOID METABOLISM (SIGAME) genes. The acetylation of SGAs in cultivated and wild tomatoes is catalyzed by the BAHD-type acyltransferase GAME36, which is involved in the regulation of α -tomatine biosynthesis and degradation [111]. This suggests that the detoxification

pathway of SGA may be jointly regulated by multiple epigenetic modifications. Although detailed studies have been conducted on acetylation modifications involved in the production of secondary metabolites in fruits, some pathways remain unidentified. For example, the accumulation of the primary metabolite citric acid in citrus fruits correlated with low levels of DNA methylation of *PH5* [72]. Meanwhile, histone acetyltransferases (CitHAG11/28) show a predominantly negative correlation with citric acid content [112]. Is there a regulatory pathway linking these two processes? Overexpression of lysine deacetylase during polyploidization shifts metabolic flux from primary to secondary metabolism, with acetylation reprogramming playing a crucial role in this process [113]. Does this reprogramming affect the conversion of citrate into other secondary metabolites? These changes in epigenetic modifications may also suggest the existence of significant connections. Such variations may also indicate important connections in this process.

RNA modifications

RNA modification refers to the various chemical modifications that occur on RNA and are widely found in the four nucleotides A, U, C, and G. Among these, methylation modifications on RNA nucleosides constitute one of the primary forms of RNA modification, including *N*¹-methyladenosine (*m*¹A), 3-methylcytosine (*m*³C), 5-methylcytosine (*m*⁵C), *N*⁶-methyladenine (*m*⁶A), and *N*⁷-methylguanine (*m*⁷G), among others [114]. Studies have shown that RNA methylation modifications are dynamically reversible just like DNA methylation modifications, and elevating RNA modifications from a fine-tuning mechanism to a new regulatory level known as the epitranscriptome. In fact, it does not fall within the scope of classical epigenetics. Further research into the characteristics and regulatory mechanisms of RNA modifications will enhance our understanding of the biology of RNA modifications from the perspective of epitranscriptomics. During fruit ripening, research on RNA methylation modifications with *m*⁶A at its core has gradually advanced, and whole-genome RNA methylation maps for *m*¹A and *m*⁵C have been constructed. Additionally, some attention has been directed toward RNA acetylation modifications involving *N*⁴-acetylcytidine (*ac*⁴C) (Fig. 3).

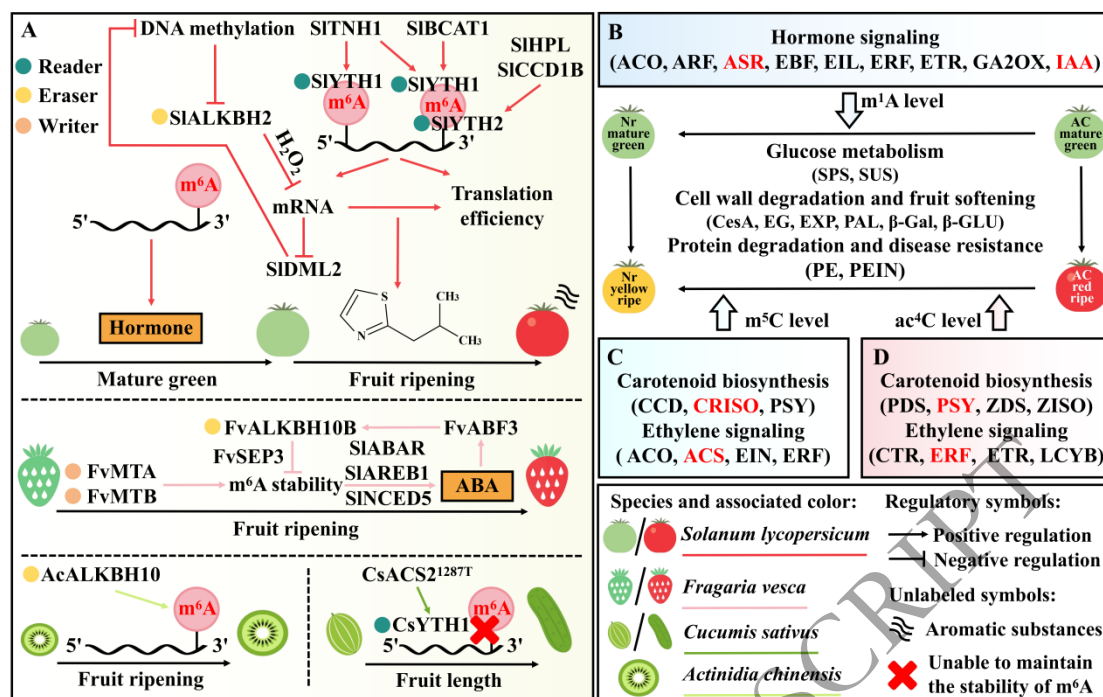


Figure 3. RNA modifications during fruit ripening. Modifications include (A) m⁶A, (B) m¹A, (C) m⁵C, and (D) ac⁴C. Fruits include tomato (*Solanum lycopersicum*), strawberry (*Fragaria vesca*), kiwifruit (*Actinidia chinensis*), and cucumber (*Cucumis sativus*). (A) "Eraser" enzymes primarily regulate fruit enlargement and ripening; "reader" enzymes mainly participate in the synthesis of aromatic volatiles and fruit length regulation. (B) The m¹A methylation levels decrease during post-veraison in *Solanum lycopersicum*, while multiple genes associated with ethylene signaling are activated. (C) Elevated m⁵C levels activate mRNAs involved in cell wall biosynthesis and secondary metabolite accumulation during the ripening of *Solanum lycopersicum*. (D) The ac⁴C modification during post-veraison significantly alters the expression of genes related to carotenoid biosynthesis, ethylene signaling, and cell wall construction in *Solanum lycopersicum* fruits. Regulatory mechanisms unique to different species are represented by lines of varying colors. Positive regulation is indicated by arrows, while negative regulation is represented by T-shaped symbols.

m⁶A modification influences fruit ripening, aroma compound release, and biological trait changes. This is the most prevalent form of methylation in eukaryotic RNA, occurring at the sixth nitrogen atom of the adenine base [115]. Chemically, this modification is highly stable, insensitive to chemical reagents, and does not affect nucleotide base pairing. With the advancement and application of high-throughput sequencing methods, three classes of methyltransferases involved in plant m⁶A modification have been progressively identified: "writers", "erasers", and "readers" [116]. "Writers" primarily regulate m⁶A deposition, including mRNA adenosine methylase (MTA),

METHYLTRANSFERASE B (MTB), FKBP12 interacting protein 37 (FIP37), VIRILIZER (VIR), and HAKAI[117]. "Erasers" represent a crucial class of demethylases, primarily mediated by the ALKB HOMOLOG (ALKBH) family in plants [118]. "Readers" possess a YTH conserved domain, with *Arabidopsis thaliana* harbouring 13 members, which is significantly more than the five found in humans [119]. Firstly, these three classes of methyltransferases involved in m⁶A modification play a critical and synergistic role during fruit ripening [120-123]. For example, in *Solanum lycopersicum*, "erasers" proteins are primarily associated with fruit enlargement and ripening; "readers" proteins are primarily associated with the synthesis of aromatic volatiles. From immature green stage to green-ripe mature stage, overall m⁶A levels increase during fruit enlargement in *Solanum lycopersicum*, with thousands of protein-coding genes undergoing m⁶A modification in 3' untranslated regions (UTR) [124]. This is primarily regulated by SlALKBH2, which is sensitive to H₂O₂-mediated oxidative modifications. SlALKBH2 promotes protein stability by forming homodimers via intermolecular disulfide bonds [125]. Mutations in SlALKBH2 regulate the DNA demethylase SlDML2 and delay fruit ripening [126]. In addition, "reader" protein controls the release of aroma compounds in *Solanum lycopersicum* fruit. Hydroperoxide lyase (SlHPL) and carotenoid cleavage dioxygenase 1B (SlCCD1B), which are associated with the synthesis of volatile compounds, accumulate in SlYTH2-induced aggregates and negatively regulate fruit aroma [127]. SlYTH1 exerts an opposite effect to SlYTH2 in regulating the release of aromatic compounds. Knockout of SlYTH1 specifically reduces the content of flavor-related volatiles in fruits without affecting ripening time, ethylene production, or fruit firmness [128]. Obtaining fruits with intense aroma without compromising yield and quality may represent a key target for future molecular breeding efforts. Significant m⁶A modifications in *Actinidia chinensis* also occur at stop codons and in 3'UTR, with m⁶A levels gradually decreasing during fruit ripening. Among these, AcALKBH10 plays a crucial regulatory role in soluble sugar accumulation [129]. Furthermore, a recent study revealed a dominant interaction between YTH1 and ACS2. In wild cucumber (*Cucumis sativus* var. *hardwickii*), ACS2^{1287C} mutation triggered the addition of an m⁶A modification to a neighbouring adenosine residue, which was recognised by YTH1. The equilibrium of RNA folding was shifted towards the weakest structural conformation, which resulted in shorter fruit. In contrast, in cultivated cucumber (*Cucumis sativus*), ACS2^{1287T} mutation disrupted m⁶A methylation, formed a more compact RNA conformation and resulted in elongated fruit [130]. This discovery provides crucial genetic evidence that epitranscriptomic regulation shapes biological traits. Secondly, m⁶A

modification primarily affects mRNA stability during fruit ripening. In *Solanum lycopersicum*, SLALKBH2 primarily inhibits fruit ripening by reducing the mRNA abundance of the DNA demethylase SLML2 [126]. During the onset of fruit ripening in *Fragaria vesca*, m⁶A modifications in coding sequence (CDS) regions tend to stabilise mRNA, whereas m⁶A at stop codons and in 3'UTR shows a negative correlation with the abundance of associated mRNA. FvMTA and FvMTB are indispensable in this process, as MTA-mediated m⁶A modification enhances the mRNA stability of NINE-CIS-EPOXYCAROTENOID DIOXYGENASE 5 (FvNCED5) and ABSCISIC ACID RESPONSIVE ELEMENTS-BINDING PROTEIN (FvAREB1) [131]. Finally, m⁶A modification is involved in the hormonal signaling pathways that regulate fruit ripening. Studies in *Solanum lycopersicum* have shown that overall m⁶A levels increase during fruit enlargement regulated by SLALKBH2, and genes associated with the ethylene signaling pathway are activated and expressed [124, 125]. In typical non-climacteric fruits *Fragaria vesca*, "writer" and "eraser" proteins participate in ABA signalling pathway through positive and negative regulation [131]. FvALKBH10B is also activated by ABA and positively regulates the stability of SEPALLATA3 (FvSEP3) mRNA via m⁶A demethylation. Mutations in FvALKBH10B cause delayed fruit ripening and induce hypermethylation of m⁶A in 1,859 genes [132]. Although a series of studies have investigated the regulatory mechanisms of m⁶A modification in fruits, research on its role in key secondary metabolite biosynthesis and color-related substance changes remains insufficient, with limited fruit types studied. This is primarily constrained by the difficulty in obtaining perennial woody fruit tree materials and the lack of stable genetic transformation techniques.

RNA modifications participate in cell wall construction and hormone signaling pathways during fruit ripening. In addition to m⁶A modifications, RNA methylation modifications (including m¹A and m³C) as well as RNA acetylation modifications (such as ac⁴C) have been identified during fruit ripening. m¹A is an RNA methylation modification occurring on the first nitrogen atom of the adenine base. This modification is primarily distributed in the first exon of coding sequences. Unlike m⁶A, m¹A-modified sequences lack distinct characteristics, exhibit GC base enrichment around modification sites, and have lower minimum free energy [133]. m¹A modification is widely present in *Solanum lycopersicum* mRNAs and can also be detected in long noncoding RNAs (lncRNAs) and circular RNAs (circRNAs), which are primarily enriched in start codons and coding regions. From the green-ripe stage to 6 days post-veraison, the overall m¹A methylation level in fruit

decreases, a process accompanied by significant differential expression of genes associated with hormone signaling pathways [134]. m⁵C, a highly dynamic methylation modification on mRNA, shows significant enrichment at translation initiation sites in human [135] and mouse [136] cells, but has not been detected in *Arabidopsis thaliana* [137]. Compared to m⁶A, research on m⁵C is still in its infancy, and its specific regulatory mechanisms remain to be elucidated. mRNAs with elevated levels of m⁵C participate in responses to biotic/abiotic stress and cell wall metabolism during fruit ripening in Alisa Craig (AC). In Never Ripe (Nr) mutant fruit, the expression of genes associated with carotenoid biosynthesis and ethylene signalling is primarily observed, alongside the identification of lncRNA and circRNA targets bearing distinct m⁵C sites [138]. ac⁴C is a highly conserved RNA acetylation modification widely present in transfer RNA (tRNA) and ribosomal RNA (rRNA), and has also been progressively identified in mRNA [139]. The presence of ac⁴C acetylation has also been confirmed in plant mRNAs, such as those in *Arabidopsis thaliana* and *Oryza sativa* [140]. From the green-ripe stage to 6 days after color break, acetylation levels of AC and Nr showed decreasing and increasing trends, respectively, with significant differential expression of genes related to carotenoid biosynthesis, ethylene signaling, and cell wall construction. Among these, the ac⁴C levels of key genes SUS, aconitase (ACO), ethylene response (ETR), and cellulose synthase (CesA) were down-regulated, while the ac⁴C level of β -galactosidase (β -Gal) was up-regulated. Correspondingly, lncRNA also exerted positive regulatory effects on target gene expression [141]. Since these RNA modifications have all been discovered and studied in recent years, research on fruit ripening has primarily focused on mapping genome-wide RNA modification profiles. These emerging RNA modifications may resolve existing biological questions and reveal novel regulatory mechanisms in the future.

Applications of epigenetic modification inhibitors

Unlike DNA variations such as mutations, epigenetic modifications are reversible and can be modulated by approaches such as exogenous chemical treatment, a strategy known as "epigenetic therapy". Currently, multiple epigenetic modification inhibitors have been approved for clinical trials and disease treatment [142]. During fruit ripening, DNA methylation inhibitors (5-aza-2'-deoxycytidine, 5-aza-2dC; 5-azacytidine, 5-azaC), histone modification inhibitors (Trichostatin A, TSA; Sodium butyrate, NaB; Sirtinol; 5'-deoxy-5'-methylthioadenosine, MTA), and RNA modification inhibitors (3-deazaneplanocin A, DA; meclofenamic acid, MA) have been applied (Table 1).

5-azaC is a base analog that reduces genomic DNA methylation levels by inhibiting methyltransferase activity. Low concentrations of 5-azaC effectively reduce DNA methylation levels by decreasing intracellular DNA methyltransferase activity, while high concentrations participate in RNA or DNA synthesis and exert cytotoxic effects [142]. Exogenous 5-azaC treatment significantly reduces genome-wide DNA methylation during the ripening process of fruits. Application methods for this substance include surface coating [150], spray application on fruit clusters [148], and fruit injection [104]. Exogenous application of 5-azaC to climacteric fruits promotes ripening, increases anthocyanin content, and enhances the accumulation of substances related to disease-resistance [143-145]. Exogenous application of 5-azaC to non-climacteric fruits induces premature ripening in *Vitis labruscana*, with significant expression of genes associated with fruit softening [146, 147]. This contrasts with the opposite effect observed in *Fragaria × ananassa* fruits [104]. Furthermore, this DNA methylation inhibitor affected secondary metabolite accumulation [51, 69] in citrus. In addition, 5-aza-2'-deoxycytidine (5-aza-2dC) has been found to promote anthocyanin accumulation in the peel of *Malus domestica* [68]. It is a 2'-deoxycytidine analogue that exhibits lower toxicity than 5-azaC [142]. TSA, a histone deacetylase inhibitor, is used to regulate gene expression at the transcriptional level. It induces a series of biological effects, including chromatin remodeling through protein hyperacetylation, induction of cell differentiation, and modulation of transcription factor activation/inhibition [151]. Exogenous application of TSA in *Vitis labruscana* delayed the depletion of organic acids and soluble sugars in fruit, whereas 5-azaC treatment produced the opposite effect; both inhibitors suppressed linalool release, reducing the musky aroma of fruit [148]. Furthermore, exogenous TSA application in *Vitis vinifera* affected anthocyanin accumulation [96], a finding mirrored in fruits treated with other histone deacetylase inhibitors (NaB, Sirtinol) [113, 149]. In contrast, the histone methyltransferase inhibitor MTA primarily influenced fruit ripening processes [88]. An overall increase in m⁶A levels, primarily regulated by m⁶A modifications occurring in the 3'UTR through the action of "writer" and "eraser" proteins, is frequently observed during fruit enlargement in *Solanum lycopersicum*. Injecting fruit with inhibitors of "writers" (DA) and "erasers" (MA) of m⁶A inhibits fruit enlargement, while DA simultaneously promotes fruit ripening [124]. This indicates that fruit ripening may be impacted by various epigenetic alterations, and that these effects vary in aspects such as physical development, color changes, and the emission of aromatic substances. In addition, some of the epigenetic modifiers currently in use exhibit a certain degree of toxicity. Among these, 5-azaC, which is widely

used in plant research, is particularly toxic and carcinogenic [152]. Are they truly suitable for regulating fruit ripening via injection or soaking? On the one hand, this could raise consumer concerns about food safety. On the other hand, these epigenetic modification inhibitors are all quite expensive, which also significantly increases agricultural production costs.

Table 1. The effects of epigenetic modification inhibitors in fruits.

Species	Nam es	Types	Functions	Regulatory relationship	Refer ences
<i>Malus domestica</i>	5- aza- 2dC	DNA methylation inhibitor	Increased anthocyanin content in fruit.	Negative correlation between DNA methylation and gene expression in the flavonoid pathway.	[68]
<i>Solanum lycopersicum</i>	5- azaC	DNA methylation inhibitor	Promoted early fruit ripening; Reduced carotenoids; Increased the accumulation of ascorbic acid in fruit; Reduced susceptibility to gray mold.	Decreased DNA methylation levels; RIN transcription factors bind to the promoters of genes associated with fruit ripening; SlGalUR5 involved in ascorbic acid accumulation; Increased expression of <i>SIDCL2</i> .	[143- 145]
<i>Vitis labruscana</i> (<i>Vitis labrusca</i> × <i>Vitis vinifera</i>)	5- azaC	DNA methylation inhibitor	Promoted fruit ripening.	Decreased DNA methylation levels; Genes associated with fruit ripening, photosynthesis, protein phosphorylation, and heat stress are regulated.	[146, 147]
<i>Citrus sinensis</i> Osbeck	5- azaC	DNA methylation inhibitor	Inhibited fruit peel yellowing.	Decreased DNA methylation levels.	[51]
<i>Prunus persica</i>	5- azaC	DNA methylation inhibitor	Increased the anthocyanin content of fruit pulp.	Decreased DNA methylation levels; Genes regulating anthocyanin biosynthesis are suppressed.	[70]
<i>Fragaria</i> × <i>ananassa</i>	5- azaC	DNA methylation inhibitor	Reduced carotenoid and anthocyanin contents; Inhibited fruit ripening.	Decreased DNA methylation levels.	[104]

<i>Pyrus bretschneideri</i>	5-azaC	DNA methylation inhibitor	Increased levels of abscisic acid, β -carotene, and lutein in fruit pulp.	Decreased DNA methylation levels; Increased expression of ABA biosynthesis genes.	[65]
<i>Citrus sinensis</i> (Linn.) Osbeck	5-azaC	DNA methylation inhibitor	Increased the anthocyanin content of fruit pulp.	Decreased DNA methylation levels; Reduced levels of DNA methylation in the promoter regions of genes associated with anthocyanin synthesis and accumulation.	[69]
<i>Vitis labruscana</i> (<i>Vitis labrusca</i> $\times$ <i>Vitis vinifera</i>)	5-azaC TSA	DNA methylation inhibitor Histone deacetylase inhibitor	TSA delayed the depletion of organic acids in post-harvest grapes, whereas 5-azaC had the opposite effect.	<i>VvHDA15</i> affected fruit quality after harvest through acetylation modification.	[148]
<i>Vitis vinifera</i>	TSA	Histone deacetylase inhibitor	Increased the anthocyanin content of fruit pulp.	<i>VvERF4</i> interacted with <i>VvHDAC19</i> , and together downregulated the expression of target genes involved in anthocyanin biosynthesis.	[96]
<i>Citrus clementina</i> hort. (<i>Citrus paradisi</i> $\times$ <i>Citrus reticulata</i>)	NaB Sirtinol	Histone deacetylase inhibitor	Increased the anthocyanin content.	The inhibitors did not inhibit the expression level of <i>CgSRT2</i> and <i>CgHDA8</i> .	[113]
<i>Vitis labruscana</i> (<i>Vitis labrusca</i> $\times$ <i>Vitis vinifera</i>)	Sirtinol	Histone deacetylase inhibitor	Increased the anthocyanin content of fruit peel.	Post-translational modifications occurred in anthocyanin biosynthesis-related genes.	[149]
<i>Vitis vinifera</i>	MTA	Histone methyltransferase inhibitor	Promoted early fruit ripening.	<i>VviAGL6a</i> directly bound to <i>VviJMJ21</i> promoter.	[88]
<i>Solanum lycopersicum</i>	DA MA	m ⁶ A writer inhibitor m ⁶ A eraser inhibitor	Inhibited fruit enlargement; DA promoted fruit ripening; MA inhibited fruit ripening.	The m ⁶ A modification genes involved in fruit enlargement were regulated by DA or MA.	[124]

Application of epigenetic modification sequencing technology

With the continuous advancement of epigenetic research, numerous sequencing technologies for epigenetic modifications have been developed. Based on varying detection requirements and sample types, epigenetic modification sequencing has progressively expanded from single-gene loci to multi-gene loci, and ultimately to genome-wide loci. Using fruits as sequencing materials, comprehensive epigenetic modification sequencing has been performed across DNA, histone, and RNA levels, significantly advancing research into fruit quality and ripening processes (Table 2).

Whole-genome bisulfite sequencing (WGBS) technology combines bisulfite treatment with high-throughput sequencing to detect individual cytosine methylation levels across the entire genome. It is currently the most widely applied DNA methylation sequencing technique in plants [156]. This method enables high-coverage screening of unknown DNA methylation regions in samples, including low CpG density areas. Extensive WGBS analyses have been conducted on fruits, covering major cultivated fruit species worldwide. However, fruit ripening is often accompanied by tissue specificity and varying growth rates. Currently, the application of sequencing technologies for DNA methylation analysis largely relies on the whole-genome level rather than specific regions, which limits the utility of the resulting epigenetic modification profiles and may lead to inconsistencies with transcriptomic data. Additionally, a combined analysis of WGBS and Chromatin Immunoprecipitation Sequencing (ChIP-seq) has been conducted in *Solanum lycopersicum*, enabling systematic and in-depth data mining of regions modified by H3K4me3, H3K27me3, Histone H3 lysine 27 acetylation (H3K27ac), and histone H3 lysine 9 acetylation (H3K9ac) [73]. ChIP-seq is a research method that combines ChIP technology with next-generation sequencing to reveal DNA-protein interactions. It has been widely applied in studies of RNA polymerase [157], histone modifications [158], and the mechanisms underlying transcription factor function and regulation [59], playing a significant role in investigating gene expression in eukaryotes. In *Solanum lycopersicum* [60, 85, 92], *Malus domestica* [95], and *Pyrus pyrifolia* × *Pyrus communis* [109] fruits, ChIP-seq technology has been employed to analyze specific histone modification regions, particularly those involving histone H3. Furthermore, this sequencing technology has begun to be integrated with other omics techniques for combined omics analysis, such as Chromatin State Analysis [97] and DNase I Hypersensitive Sites Sequencing (DNase-seq) [153]. The combination of multi-omics approaches with genome-wide or site-specific analyses has largely overcome the limitations imposed by the unique characteristics of the material or the scope

of detection. However, this approach remains underutilized in studies of fruit ripening, with research primarily relying on ChIP-seq or ChIP-qPCR at specific sites. Methylated RNA immunoprecipitation sequencing (MeRIP-seq) is the earliest developed and most widely applied m⁶A sequencing technology. It utilizes specific antibodies to enrich RNA containing m⁶A, addressing the challenge that this modification does not affect nucleotide base pairing capacity and is insensitive to chemical reagents [115]. Currently, MeRIP-seq has become the primary technique for studying m⁶A modifications that influence fruit ripening processes and quality. In addition, techniques such as Methylated DNA immunoprecipitation sequencing (MeDIP-seq) [134, 138] and N⁴-acetylcytidine sequencing (ac⁴C-seq) [141], which are used to study RNA methylation and acetylation, respectively, have recently been applied. These approaches may also emerge as important methods for addressing existing biological questions in the future. Overall, multi-taxon and multi-omics population studies have become the primary approach for elucidating plant evolution and molecular regulatory mechanisms in the post-genomic era. Integrating epigenomic sequencing with other omics technologies has emerged as a key strategy for deciphering fruit ripening processes and quality changes.

Table 2. Application of sequencing technologies associated with epigenetic modifications in fruits.

Species	Sequencing technologies	Analysis types	Analysis scope	References
<i>Solanum lycopersicum</i>	WGBS ChIP-seq	DNA methylation	Whole-genome, H3K4me3, H3K27me3, H3K27ac, H3K9/K14ac	[59, 73, 74]
<i>Fragaria × ananassa</i>	WGBS	DNA methylation	Whole-genome	[53]
<i>Citrus sinensis</i> Osbeck	WGBS	DNA methylation	Whole-genome	[51]
<i>Malus domestica</i>	WGBS	DNA methylation	Whole-genome	[66, 68]
<i>Citrus limon</i>	WGBS	DNA methylation	Whole-genome	[71]
<i>Pyrus pyrifolia</i>	WGBS	DNA methylation	CG and CHG methylation	[64]
<i>Pyrus bretschneideri</i>	WGBS	DNA methylation	Whole-genome	[65]
<i>Vitis vinifera</i>	WGBS	DNA methylation	Whole-genome	[50, 53]
<i>Prunus persica</i>	WGBS	DNA methylation	Whole-genome	[58]
<i>Citrus sinensis</i> L. Osbeck	WGBS	DNA methylation	Whole-genome	[69]
<i>(Vitis bryoniifolia × Vitis vinifera) × Vitis vinifera</i>	WGBS	DNA methylation	Whole-genome	[67]

<i>Vitis labrusca</i> × <i>Vitis vinifera</i>	WGBS	DNA methylation	CHH hypermethylation	[52]
<i>Solanum lycopersicum</i>	ChIP-seq	Histone modification	H3K4me3, H3K9Ac and H3K27Ac	[60, 85, 92]
<i>Vitis vinifera</i>	ChIP-seq	Histone modification	H3K4me3	[89]
<i>Malus domestica</i>	ChIP-seq	Histone modification	H3K9ac	[95]
<i>Pyrus pyrifolia</i> × <i>Pyrus communis</i>	ChIP-seq	Histone modification	H3	[109]
<i>Fragaria vesca</i>	ChIP-seq Chromatin state analysis	Histone modification	H3K9/K14ac, H3K27ac, H3K4me3 and H3K36me3	[97]
<i>Musa acuminata</i>	DNase-seq ChIP-seq	Histone modification	H3K4me3, H3K9ac, H3K27ac and H3K27me3	[153]
<i>Solanum lycopersicum</i>	MeRIP-seq MeDIP-seq ac ⁴ C-seq	RNA modifications	m ⁶ A modifications m ¹ A modifications RNA m ⁵ C methylation ac ⁴ C methylation	[124-126, 128, 134, 138, 141, 154]
<i>Fragaria vesca</i>	MeRIP-seq	RNA modifications	m ⁶ A modifications	[129, 130]
<i>Actinidia chinensis</i>	MeRIP-seq	RNA modifications	m ⁶ A modifications	[131, 155]
<i>Cucumis sativus</i>	MeRIP-seq	RNA modifications	m ⁶ A modifications	[132]

Conclusion

Research on DNA, histone, and RNA modifications has significantly advanced our understanding of the processes and quality changes involved in fruit ripening, particularly with regard to elucidating the molecular regulatory networks. However, studies on epigenetic modifications during fruit ripening continue to present opportunities and challenges. A comprehensive strategy is therefore required to investigate these modifications (Fig. 4). Firstly, combine epigenetic modifications during fruit ripening with multi-omics technologies, such as genomic sequencing represented by Telomere-to-telomere (T2T), high-throughput/resolution chromosome conformation capture (Hi-C), high-Fidelity Sequencing (Hi-Fi), and Micrococcal Nuclease-based Chromosome Conformation Capture (Micro-C); chromatin conformation studies represented by Assay for Transposase-Accessible Chromatin with high-throughput sequencing (ATAC-seq) and chromatin state analysis; high-resolution histone sequencing studies represented by Cleavage Under Targets and Tagmentation (CUT&Tag) and Cleavage Under Targets and Release Using Nuclease

(CUT&RUN); and single-cell and spatial omics studies represented by single-cell transcriptome (scRNA-seq), spatial enhanced resolution omics sequencing (Stereo-seq), spatial transcriptomics RNA sequencing (STRNA-seq), and spatial proteomics. The introduction of these technologies will significantly enhance our understanding of the deeper epigenetic modifications and their corresponding regulatory relationships during fruit ripening. Secondly, elucidate the profound relationships among DNA, histone, and RNA modifications. Future research should focus on interpreting the tentative findings and uncovering the unidentified regulatory roles identified in these preliminary investigations. For example, the deep synergies between epigenetic modifications and transcriptional activation, mRNA stability, or hormone signaling; how changes in DNA methylation affect chromatin accessibility; whether histone acetylation and methylation events precede or accompany DNA demethylation; and mechanisms by which epigenetic modifications influence the biosynthesis pathways of secondary metabolites by regulating key structural genes. New epigenetic editing tools (such as CRISPR-dCas9-based DNA demethylation or histone modification systems) may help address these critical biological questions. This approach can help improve fruit development processes and quality through the regulation of epigenetic modifications, and drive innovation in molecular breeding. Finally, it is crucial to clarify the advantages and disadvantages of applying epigenetic modification inhibitors, which are a "double-edged sword". If applied externally via injection, immersion, or topical application, could toxic residues remain in the fruit? Are there safer, effective alternatives that can achieve these changes in epigenetic modification levels? Addressing these questions will significantly improve the feasibility of translating epigenetic modification research from the laboratory to practical agricultural production.

A comprehensive strategy integrating studies on multiple epigenetic modifications is crucial for addressing current limitations. This approach will clarify unknown molecular regulatory mechanisms while enabling the application of these novel technologies to regulate fruit ripening timing, improve postharvest preservation, and extend shelf life in horticultural fruit plants. By focusing on these understudied areas, we will further advance our understanding of fruit ripening processes and refine existing integrated regulatory strategies.

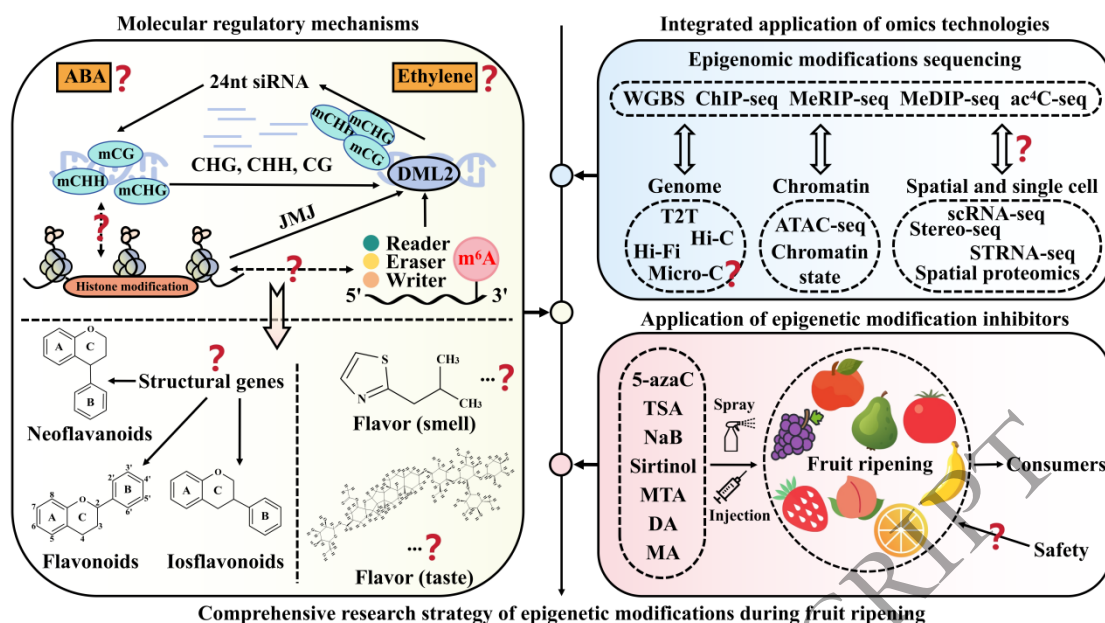


Figure 4. A comprehensive strategy model for research on epigenetic modifications during fruit ripening. It comprises three components: the integrated application of omics technologies, the exploration of molecular regulatory mechanisms, and the application of epigenetic modification inhibitors. Firstly, combine epigenetic sequencing technologies with novel sequencing techniques at the genomic, chromatin, spatial, and single-cell levels to map the epigenetic modifications influencing fruit ripening through multi-omics integration analysis. Secondly, based on these multi-omics epigenetic maps, focus on deciphering the unknown regulatory relationships among DNA, histone, and RNA modifications during fruit ripening. Clarify how epigenetic modifications influence the ripening process of both climacteric and non-climacteric fruits via ethylene and ABA signaling pathways. Further investigate the effects of epigenetic modifications on changes in fruit secondary metabolism, as well as previously undiscovered regulatory relationships involving structural genes. Finally, evaluate the impacts of exogenous epigenetic modification inhibitors on food safety and cultivation costs, and screen for harmless regulatory systems for fruit ripening that are suitable for agricultural production. The areas marked with "?" represent currently unresolved issues in the study of epigenetic modifications during fruit ripening.

Acknowledgements

This work was supported by the Zhejiang Provincial Natural Science Foundation of China – Hangzhou Joint Fund (Grant No. LHZSD24C150001), the National Natural Science Foundation of China (Grant No. U25A20688), and the Science and Technology Program of Sichuan Province (Grant No. 2026YFHZ0239).

Author contributions

T. Z. and C. S. conceived and designed this review article. T. Z., C. Z., F. X., and Z. Y. completed literature organization. T. Z. and C. S. drafted the manuscript, wrote the review, made the tables, and drew the images. T. Z., C. Z., F. X., Z. Y., Y. C., J. C., S. Y., H. H., and C. S. participated in discussions and provided some valuable advice. S. L., J. C., and Y. W. helped to revise the manuscript. All authors were involved in reviewing and editing the manuscript, and approved the final manuscript.

Data availability

All data are available in the tables and figures of the manuscript.

Conflict of interest statement

The authors declare that they have no competing interests.

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